

Utility-Peak Itemset Mining with Constraint Programming

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This is a 2-minute summary — come see our poster for the full story!

Introduction — What is Data Mining?

Finding hidden patterns in large collections of data.

- ▶ Retail: which products are bought together?
- ▶ Medicine: which symptoms co-occur?
- ▶ Finance: which transactions are anomalous?

Our goal: discover the most profitable patterns hidden in transactional data.

Example — Transactional Database

TID	A	B	C	D	E
T ₁	1		2	1	
T ₂	3	1			2
T ₃		2	1	4	1
T ₄	1		3		2

1. Motivation — High-Utility Itemset Mining (HUIM)

Finds patterns that are truly valuable — profit $\times$ quantity, not just how often items appear together.

- Example: {CD} has utility 18 ✓ — more profitable than {ACD} with utility 9 ✗

Limitations of HUIM

- Utility is non-monotone — no Apriori-style pruning applies.
- At low thresholds: output explodes into millions of HUIs — impossible to use.

Need: a compact, utility-based condensed representation.

Example — Transactional Database

TID	A	B	C	D	E
T ₁	1		2	1	
T ₂	3	1			2
T ₃		2	1	4	1
T ₄	1		3		2
Profit	4	2	1	3	2

2. New Condensed Representation — Utility-Peak Itemsets (UPIs)

A UPI = a local utility maximum in the itemset lattice.

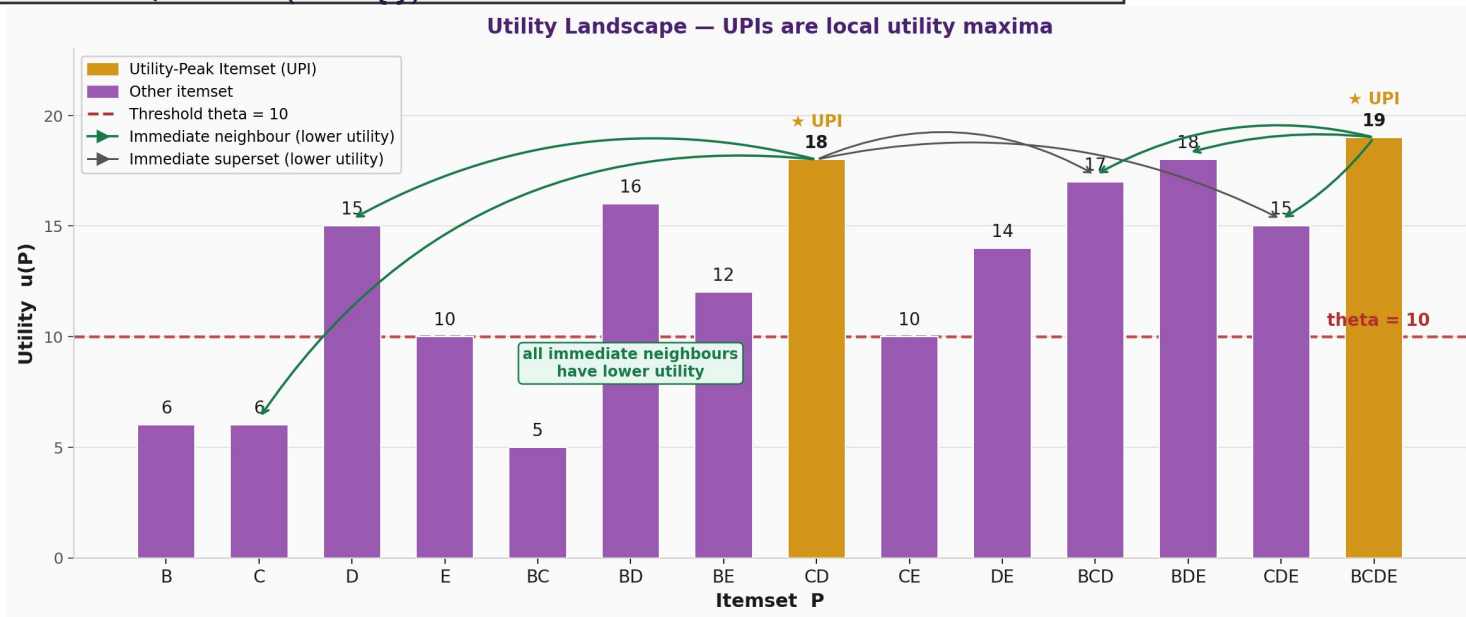
**P is a UPI
iff:**

$$\forall i \in P : u(P \setminus \{i\}) \leq u(P)$$
$$\forall i \notin P : u(P \cup \{i\}) \leq u(P)$$

removing any item does not improve utility

adding any item does not improve utility

✓ Much more compact and actionable than the full HUI set.



→ Full details on the poster

3. PeakUtility: A Global CP Constraint

PeakUtility Global Constraint

PeakUtility is a parametric global constraint that filters candidate itemsets to retain those satisfying the minimum utility threshold and, in UPI mode, additionally enforces local utility maximality, where neither adding nor removing a single item can increase utility.

mode = HUI

mode = UPI ★

- ▶ 8 propagation rules eliminate non-peak candidates directly during search.
- ▶ NP-hard to enforce full consistency → approximate propagation (sound, not complete).

→ *Full details on the poster*

4. Experimental Evaluation

52 datasets · Choco-Solver 4.10 · Timeout = 900 s · Baselines: EFIM, CP4HUIM

- ▶ PeakUtility competitive with EFIM on tractable instances.
- ▶ Robust where EFIM runs out of memory on large and dense datasets.
- ▶ Direct UPI mining — avoids exhaustive HUI enumeration entirely.

Full tables are on the poster.

→ *Full details on the poster*

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Problem

HUIM generates millions of patterns — impossible to use

UPIs

A compact set of patterns that are locally optimal in terms of utility.

PeakUtility

One global CP constraint · 8 rules

Results

52 datasets · competitive with EFIM · robust on large data